



Neural Networks and Deep Learning

Lecture *Machine Learning* on 10.3.2026

Foundation Models

Definition

Training

Tasks

Positionwise Classification

Data Comparison DNA to

NLP

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Foundation Models

- large-scale machine learning (ML) models trained on massive datasets
- capable of performing a range of different tasks
- examples:
 - GPT, Gemini
 - Stable Diffusion, SAM (images)
 - Whisper (speech recognition)
 - models for *biological sequences*
 - protein language models (pLMs)
 - genome foundation models (gFM)



Foundation Model for Sequential Data (Informal Definition)

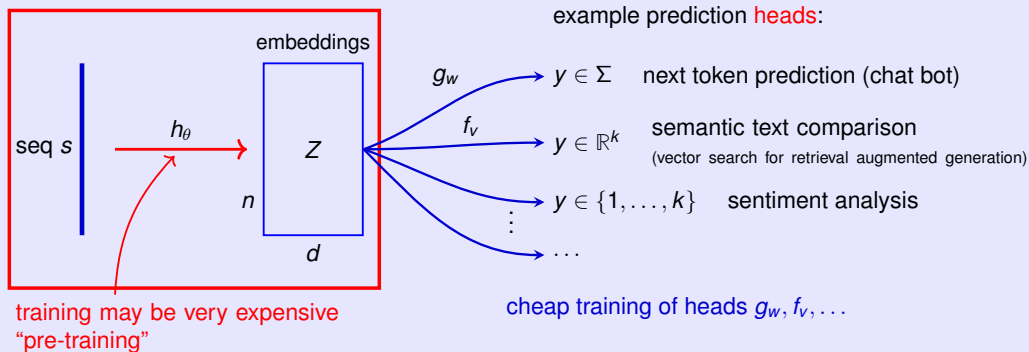
A foundation model (FM) defines an **encoder** function h_θ with parameters θ that maps sequences/texts $s = (s_1, \dots, s_n) \in \Sigma^n$ of arbitrary length n to a same-length sequence of vectors:

$$(z_1, \dots, z_n) = h_\theta(s).$$

Here, Σ is an alphabet (of characters or tokens), the *latent space dimension* d of the vectors is a model choice, the z_i are called **embeddings** and can be summarized in a matrix

$$Z \in \mathbb{R}^{n \times d}.$$

Foundation model concept



A FM is a foundation to solve diverse tasks

- Many machine-learning **tasks** are considered that require some 'understanding' of the inputs s .
- The FM h_θ learns about sequences s from a broad distribution of examples in a **task-agnostic** way and provides the embeddings Z that are input to other ML functions (prediction **heads** $g_w, f_v, \dots$) that solve individual tasks with **task-specific models and parameters**.



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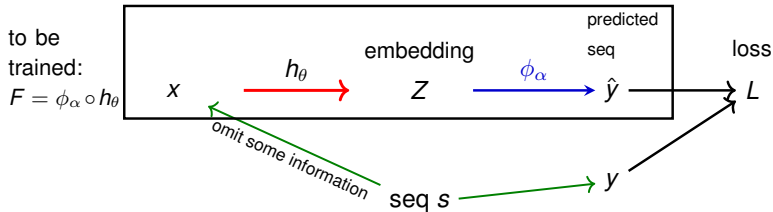
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Training Methods

- Foundation models are trained using **self-supervised learning** objectives.
- Supervision** means that target values y for a prediction are used to estimate parameters.
- Self-supervision** means that target values y and inputs x to the model are **constructed from the sequences s themselves**.
- Parameters θ and α are **trained**, i.e. chosen approximately to minimize the average **loss** $L = L(y, \hat{y})$ on a large training set of sequences s .
The loss function is chosen so that L is small when y and $\hat{y}$ are similar.





1. Unidirectional / Causal / Autoregressive Modeling (GPT-style)

The model F predicts the next character in the sequence and is a model of the conditional distribution

$$P(S_i \mid S_1, \dots, S_{i-1}).$$

Example: $s = \text{cttatgtgaaa}$

$$i = 5$$

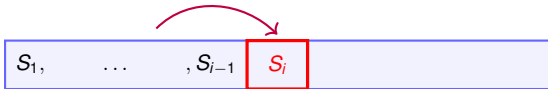
$$x = \text{ctta}$$

$$y = \text{t}$$

$$\hat{y} = P(S_i = \cdot \mid (S_1, \dots, S_{i-1}) = \text{ctta}) \in [0, 1]^4$$

Advantages:

- ① $\hat{y}$ and therefore loss values can be computed **efficiently** for all positions i in s in one run of a suitable model F , e.g., with so-called *causal attention*.
- ② F can be directly used to sample (generate) sequences (chatbot, generative/synthetic biology).



2. Bidirectional Modeling (BERT-style)

The model F predicts characters at random positions i given the *other characters* and is a model of the conditional distribution

$$P(S_i \mid S_{-i}).$$

Example: $s = \text{cttatgtgaaa}$

$i = 5$

$x = \text{ctta?gtgaaa}$

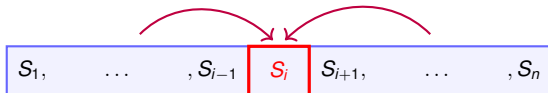
$y = \text{t}$

$\hat{y} = P(S_i = \cdot \mid (S_1, \dots, S_{i-1}) = \text{ctta}, (S_{i+1}, \dots, S_n) = \text{gtgaaa}) \in [0, 1]^4$

Advantages:

- ① More natural model for inherently undirected sequences, e.g. double-stranded DNA.
- ② Easier to make good predictions $\hat{y}$.

Disadvantage: No comparably efficient way known to compute loss for all positions i (multiple passes per sequence required).



Examples of Genome Foundation Models

Causal: Evo2

Nguyen et al. (2025). "Genome modeling and design across all domains of life with Evo 2."

- uses fast convolutions via Fourier transform (StripedHyena architecture)
- **Parameters:** 40B
- **Context Window:** 1Mb (Evo 2)
- **Training Data:** trillion characters from domains of life (Bacteria, Archaea, Eukaryota, Viruses), 15,032 eukaryotic genomes + > 28K prok. genomes + mitochondria, >2M bacteriophage genomes, chloroplasts, mRNA, and more, OpenGenome2
- **Training Resources:** 2,048 NVIDIA H100 GPUs for several months

Bidirectional: Nucleotide Transformer (NT)

Dalla-Torre et al. (2023). "The Nucleotide Transformer: Building and Evaluating Robust Foundation Models for Human Genomics."

<https://github.com/instadeepai/nucleotide-transformer/>

- transformer architecture (BERT)
- **Parameters:** 50M to 2.5B parameters
- **Context Window:** 1 million (NT v3)
- **Training Data:** OpenGenome2 (see Evo2)
- **Training Resources:** Trained on TPU v4 pods (InstaDeep^a/Google collaboration)

^adaughter of BioNTech



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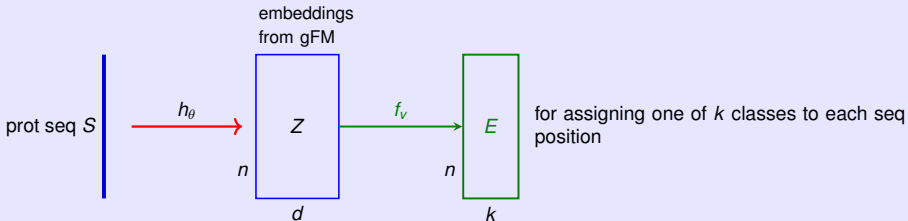
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Task: Positionwise Classification

Consider a **head** $f_v : \mathbb{R}^{n \times d} \rightarrow \mathbb{R}^{n \times k}$ (example from exercise: $d = 768$, $k = 5$)



- $E[i, j]$ could be the probability that sequence position i belongs to class j .
- Note that $E[i, \cdot]$, the i -th row of E , depends not only on character $S[i]$ but usually on the whole sequence as context.

Comparison of Training Resources

Natural Languages

- Text data set sizes for unsupervised pre-training of LLMs, not including synthetic data.
 - Llama 4: 30 trillion
 - DeepSeek V3 14.8 trillion tokens
 - Grok3 12.8 trillion tokens
- quality is an issue
- high-quality text on the internet is approximately 15 trillion tokens
- 1 token is ~ 4 characters or $\sim 3-4$ bytes of information entropy
- 15 trillion tokens correspond to 45-60 TB (tera bytes)

DNA

- in December 2025 GenBank and WGS had $6651459875408 + 42125323988215 \approx 48.8$ trillion nucleotides in **assembled genomes**, the **doubling time** was **36 months**
- the Sequence Read Archive (SRA) has 88.5 PB (peta bytes) in (unassembled) reads
- Biological sequences are strictly quality controlled: up to (rare) sequencing errors, they have been experimentally tested to code for a functional organism.